



Bicycle Design Challenge 2025-26



Design Report - 4



Team Name	Pegasus
Team ID	BDC2025020

NAME	TEAM COMPOSITION
Bhuvanan M	Captain
Kusumarekha M	Vice captain
Dharaneshwaran B	Designer-1
Ambika B	Designer-2
Aravinth R	Driver
Keerthana M	Co-Driver
Dharanidharan T	Research and development

FACULTY MENTOR:

S. Narayan

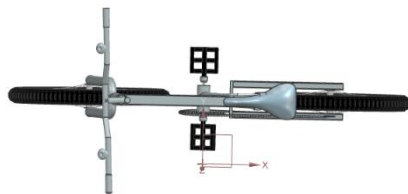
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OVERVIEW

Our Z – frame bicycle Address

The bicycle features a unique Z-frame with a minimalist structure, including front and rear wheels of equal size, a standard chain drive mechanism, and a crank set with pedals. The saddle is mounted on an inclined seat post, and the handlebars are angled for ergonomic use. A local coordinate system near the crank indicates the model used for kinematic or structural analysis.



E-BOM CREATION

The Bill of Materials (BOM) is a detailed list of all the parts, materials, and their quantities needed to make a low-cost bicycle. It helps ensure all parts are ready, making it easier to buy materials and assemble the bicycle. This BOM is made to help build an affordable and practical bicycle for today's fast-paced world.

S. No	Part. No	Description	Quantity	Material
1	ASM-001	Bicycle Frame	1	Mild Steel
2	ASM-002	Fork	1	Mild Steel
3	ASM-003	Handlebar	1	Mild Steel
4	ASM-004	Stem	1	Mild Steel
5	ASM-005	Headset Bearings	2	Steel
6	ASM-006	Seat Post	1	Mild Steel
7	ASM-007	Saddle	1	PU Foam + Cover
8	ASM-008	Crank Arm – Left	1	Mild Steel
9	ASM-009	Crank Arm – Right	1	Mild Steel
10	ASM-010	Pedal – Left	1	Plastic + Steel
11	ASM-011	Pedal – Right	1	Plastic + Steel
12	ASM-012	Chain	1	Steel
13	ASM-013	Bottom Bracket	1	Steel
14	ASM-014	Front Wheel Assembly	1	Mild Steel Rim + Rubber Tyre
15	ASM-015	Rear Wheel Assembly	1	Mild Steel Rim + Rubber Tyre
16	ASM-016	Spokes	36 × 2	Steel Wire
17	ASM-017	Front Hub	1	Steel
18	ASM-018	Rear Hub	1	Steel
19	ASM-019	Brake Set (Front)	1	Mild Steel
20	ASM-020	Brake Set (Rear)	1	Mild Steel

COST ESTIMATION AS PER E BOM:

Material Cost Calculation: Each part would have an estimated unit price that needs to be multiplied by the quantity to get the total cost for that component.

S. No	Part. No	Description	Quantity	Material	Unit Cost (INR)	Extended Cost (INR)
1	ASM-001	Bicycle Frame	1	Mild Steel	856	856
2	ASM-002	Fork	1	Mild Steel	311	311
3	ASM-003	Handlebar	1	Mild Steel	117	117
4	ASM-004	Stem	1	Mild Steel	97	97
5	ASM-005	Headset Bearings	2	Steel	117	117
6	ASM-006	Seat Post	1	Mild Steel	117	117
7	ASM-007	Saddle	1	PU Foam + Cover	195	195
8	ASM-008	Crank Arm – Left	1	Mild Steel	156	156
9	ASM-009	Crank Arm – Right	1	Mild Steel	272	272
10	ASM-010	Pedal – Left	1	Plastic + Steel	78	78
11	ASM-011	Pedal – Right	1	Plastic + Steel	78	78
12	ASM-012	Chain	1	Steel	97	97
13	ASM-013	Bottom Bracket	1	Steel	175	175
14	ASM-014	Front Wheel Assembly	1	Mild Steel Rim + Rubber Tyre	623	623
15	ASM-015	Rear Wheel Assembly	1	Mild Steel Rim + Rubber Tyre	700	700
16	ASM-016	Spokes	36 × 2	Steel Wire	350	350
17	ASM-017	Front Hub	1	Steel	136	136
18	ASM-018	Rear Hub	1	Steel	214	214
19	ASM-019	Brake Set (Front)	1	Mild Steel	156	156
20	ASM-020	Brake Set (Rear)	1	Mild Steel	156	156
Total						Rs.5001

COST REVIEW/FINALIZATION REPORT FOR TARGET PRICE ADHERENCE

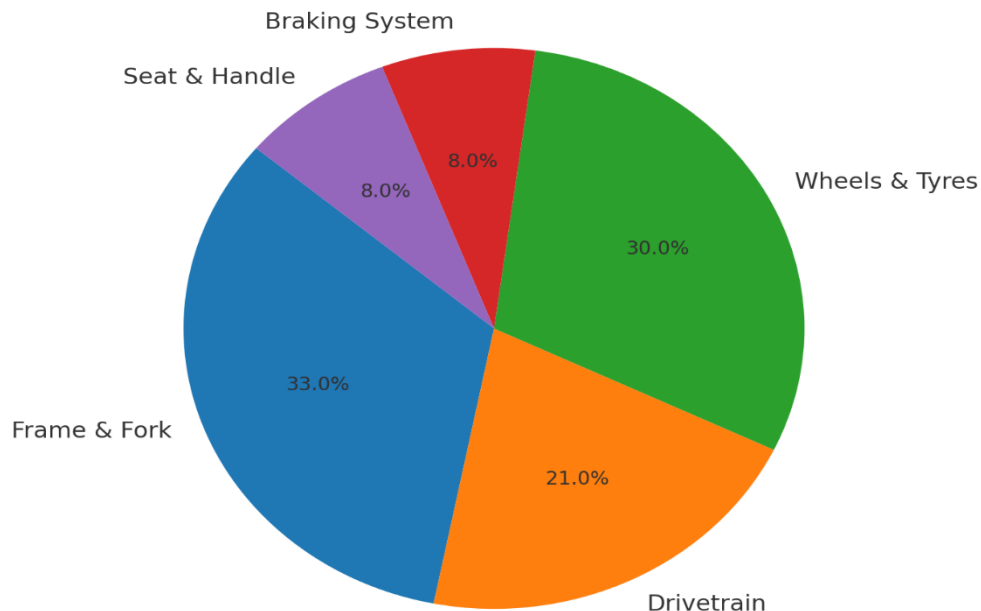
Project: Bicycle (Mild Steel)

Target Cost: ₹5,001

Reviewed By: [Team-Pegasus]

Date: [18/08/2025]

Cost Distribution of Bicycle Components (₹5,001 Total)



Cost Distribution Analysis:

- The adjusted EBOM cost is ₹5,001, which is well aligned with the defined target of ₹5,000.
- Major cost contributors are Frame & Fork (33%) and Wheels (30%).
- Drivetrain accounts for 21%, while brakes, seat, and handle components make up the remaining 16%.

Cost Optimization:

- Explore alternative suppliers for tyres and rims (high share of cost).
- Investigate use of standardized bearings and hubs to lower drivetrain expenses.

Value Engineering:

- Consider modular design changes for frame and fork to reduce fabrication costs.
- Use bulk procurement strategy for common fasteners, chains, and spokes.

Target Achievement:

- With minor optimizations, overall cost can be reduced 2–3% below target to create a buffer for price fluctuations.

DESIGN CHANGE DECISION BASED ON COST REVIEW

• Frame Material:

Explore more affordable rods for the frame that still provide enough strength and durability. Explore suppliers offering competitive rates and ensure material meets strength and fatigue requirements.

• Braking System & Wheels:

V-brakes are effective but can be optimized for cost. Evaluate cost-effective alternatives for V-brakes and wheel components without compromising performance.

- **Saddle & Stand:** Consider a simpler, budget-friendly saddle and a lightweight but economical stand. Replace premium saddles with ergonomic but simpler foam.

- **Fit and Finish Components:**

Identify low-cost options for smaller parts such as grips, bolts, and protective covers. Simplify the design of grips, pedals, bolts, nuts, and protective caps. Using standardized parts reduces both manufacturing and procurement costs.

- **Handlebars & Paint:**

Look into bulk and paint to minimize purchase costs. Consider powder coating instead of high-cost paint for durability and reduced expense.

- **Chain & Drivetrain:**

Use standard chain sets and crank sets that are easily available at lower cost. We Avoid overly customized parts that increase sourcing expenses.

- **Sustainability Consideration:**

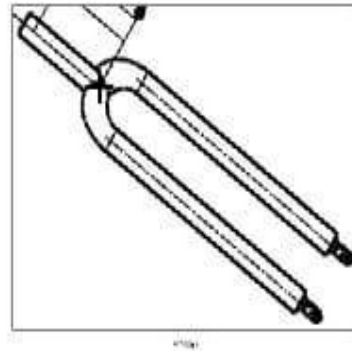
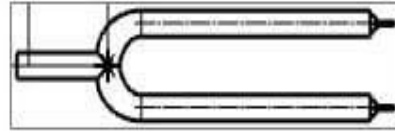
Explore recycled raw materials (like recycled Steel) for non-critical parts, reducing both cost and environmental impact.

MANUFACTURING DRAWING CREATION/RELEASE



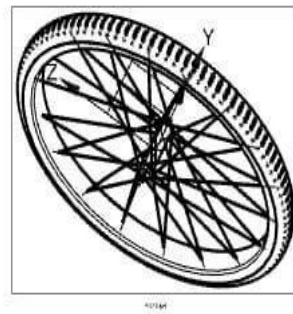
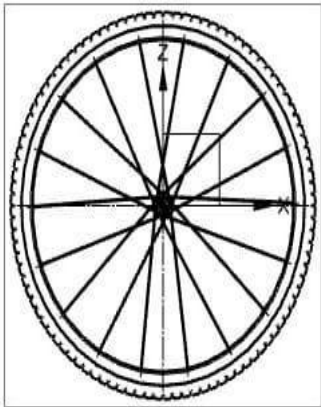
FREE HAND SKETCH

FORK OF BICYCLE



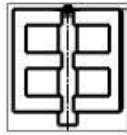
Scale 1:1	All Dimensions Are In mm
Team Number	BDC2025020
Team Name	Pegasus
College	Knowledge Institute of Technology

TYRE OF BICYCLE



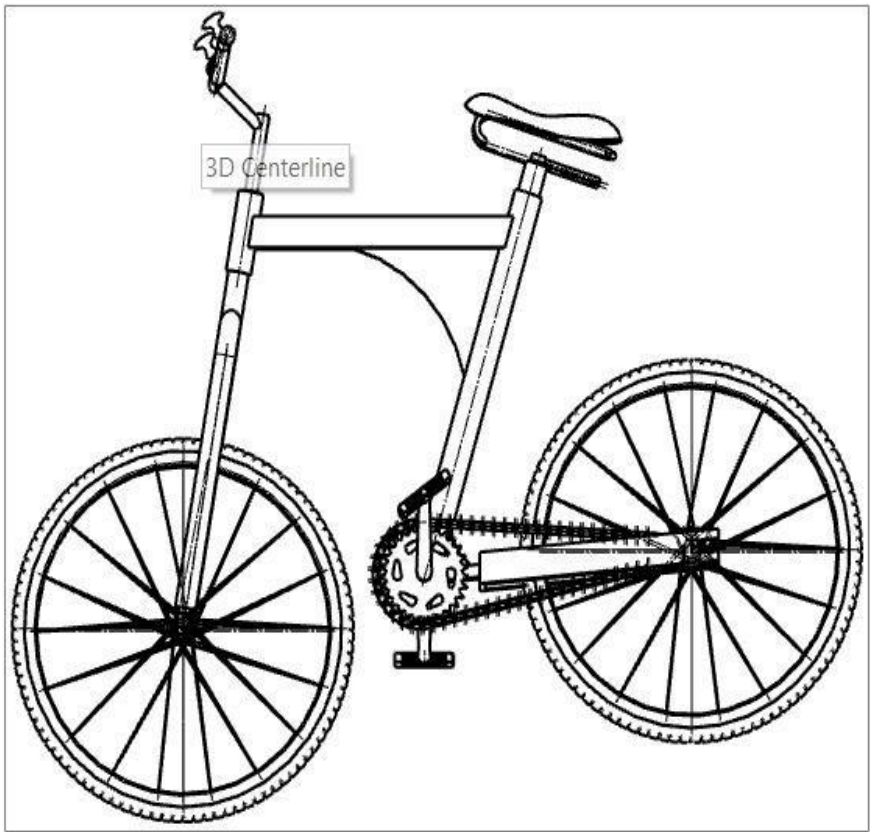
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Team Number	BDC2025020
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PEDAL OF BICYCLE



Scale 1:1	All Dimensions Are In mm
Team Number	BDC2025020
Team Name	Pegasus
College	Knowledge Institute of Technology

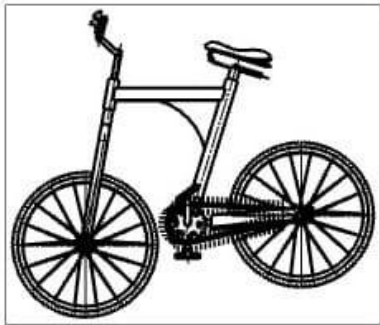
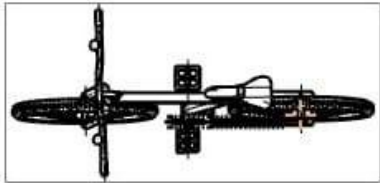
DETAILED PART DRAWING



1	WHEELSET	1
2	SEAT AND POST	1
3	FRAME AND HEADSET	1
4	HANDLEBARS	1
5	PEDESTAL	1
6	CRANKSET	1
7	REAR DERAILLEUR	1
8	REAR WHEEL	1
9	FRONT WHEEL	1
10	FRONT DERAILLEUR	1
11	SHIFTERS	2
12	SPACERS	2
13	SPACERS	2
14	SPACERS	2
15	SPACERS	2
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100	SPACERS	2

FINAL PRODUCT DISPLAY BY RENDERING/PHOTOSHOP

2D DRAWING



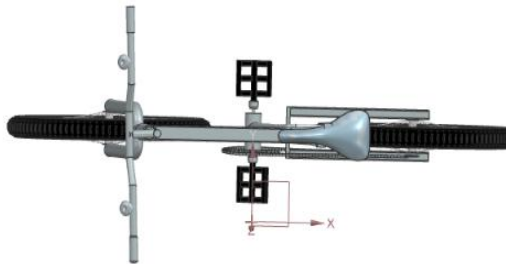
3D VIEW



FRONT VIEW



TOP VIEW



SIDE VIEW



PHOTOSHOP



Z FRAME BICYCLE - PEGASUS



CATALOGUE AND USP OF THE PRODUCT

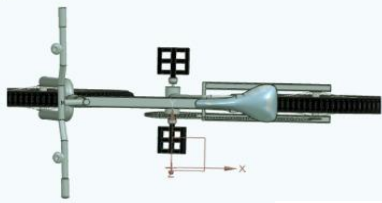
USP of the Product:

The Urban Compact Bicycle is engineered for modern riders who seek simplicity, durability, and comfort. Its robust Z-frame structure ensures stability, while the ergonomic saddle and handlebar design provide an effortless riding posture. With a compact geometry, it is ideal for urban commuting, fitness riding, and eco-friendly transport. This model prioritizes low maintenance, cost-effectiveness, and rider convenience.



INTRODUCING THE PRODUCT

CATALOGUE FOR OUR BICYCLE



FEATURES

Z - Frame Construction
Frame design ensures excellent load-bearing capacity and durability.

Ergonomic Handlebar
Optimized grip for better control and safety.

Comfortable Saddle
Rider-friendly seat with adjustable Length for different users.

Mount & Kickstand for Bicycle
The lightweight kickstand ensures stable parking, prevents damage, and adds convenience for everyday use.



FITNESS & HEALTH
Daily exercise for a healthier lifestyle



ECO-FRIENDLY RIDE
Zero emissions, sustainable transport



COMPACT STORAGE
Easy to park & store in small spaces



CONCLUSION

Our team, **Pegasus**, has developed a practical, stable, and eco-friendly bicycle featuring a Z-frame and a spring based seat suspension for enhanced comfort. Our design ensures durability, user convenience, and aligns with modern urban commuting needs.

thank you 